

A PERSONAL DATA ANALYTICS PROJECT

RETAIL SALES & FINANCIAL PERFORMANCE ANALYSIS

Retail Sales Data Analysis and Business Insights

Prepared by
Himanshu Khade

Tools and Technologies

Microsoft Excel | Microsoft SQL Server | Power BI | DAX

2026

Table of Contents

EXECUTIVE SUMMARY	3
1. PROJECT OVERVIEW	3
2. PROJECT OBJECTIVE.....	4
3. DATASET DESCRIPTION	4
3.1 Dataset Overview	4
3.2 Dataset Fields	5
3.3 Supporting Tables.....	5
3.4 Data Usage	6
4. TOOLS & TECHNOLOGIES.....	6
4.1 Microsoft Excel.....	6
4.2 Microsoft SQL Server	6
4.3 Microsoft Power BI	6
4.4 DAX	7
5. PROJECT WORKFLOW	7
6. DATA CLEANING AND EXCEL ANALYSIS	8
6.1 Data Preparation	8
6.2 Pivot Table Analysis.....	9
6.3 KPI Calculation	10
6.4 Initial Findings from Excel.....	10
7. SQL DATA ANALYSIS	9
7.1 SQL Server Database Setup	10
7.2 SQL Operations Performed	10
7.3 Sample Queries and Results	10
7.4 SQL Analysis Outcome	11
8. POWER BI DASHBOARD DEVELOPMENT	11
8.1 Data Connection and Preparation	11
8.2 Key Performance Indicators	12
8.9 DAX Measures and Calculations	12
8.10 Dashboard Design and Formatting.....	12
8.11 Final Power BI Dashboard	13
9. KEY BUSINESS INSIGHTS	14
10. LIMITATIONS AND ASSUMPTIONS	14
11. CONCLUSION	14

EXECUTIVE SUMMARY

This report documents an end-to-end analysis of 10,194 retail order lines (5,111 orders, January 2023 to December 2026) using Excel, SQL Server, Power BI, and DAX. The business generated 2.33M in sales and 292.3K in profit, a 12.6% profit margin.

4.65M Total Sales	585.1K Total Profit	77K Total Quantity	5,000 Total Orders	12.57% Profit Margin
-----------------------------	-------------------------------	------------------------------	------------------------------	--------------------------------

Headline findings

- **Growth:** sales rose from 494.0K in 2023 to 745.6K in 2026; profit rose from 51.7K to 95.9K.
- **Furniture is the weak spot:** it delivers 32.4% of sales but only 6.7% of profit (2.6% margin), against about 17% for Technology and Office Supplies.
- **Discounts drive losses:** order lines discounted above 20% lost 136.0K combined, while undiscounted lines earned a 29.6% margin.
- **Central region and several states lag:** Central earns a 7.9% margin (average discount 24.1%), and 12 states are loss-making, led by Texas (−25.7K) and Ohio (−17.0K).
- **Seasonality:** September to December produce 51.5% of annual sales; February is the weakest month.

Priority actions

- Tighten discount rules, especially above 20%.
- Review pricing and costs for Tables, Bookcases, and Supplies.
- Investigate Central and the loss-making states before expanding there.
- Plan stock and staffing for the September to December peak.

1. PROJECT OVERVIEW

Retail Sales & Financial Performance Analysis is a data analytics project developed using Microsoft Excel, Microsoft SQL Server, Power BI, and DAX to analyze a retail sales transaction dataset.

The project focuses on understanding business performance through sales, profit, quantity, orders, product, category, regional, customer segment, and monthly sales analysis.

The data was first prepared and cleaned using Excel, followed by SQL-based business analysis. The analyzed data was then connected to Power BI to develop an interactive dashboard containing key performance indicators, analytical charts, DAX calculations, and interactive filters.

2. PROJECT OBJECTIVE

The main objectives of this project are:

- To clean and prepare retail transaction data for analysis.
- To perform initial data analysis using Microsoft Excel.
- To use SQL queries for extracting meaningful business insights.
- To calculate and analyze important performance metrics such as sales, profit, quantity, orders, and profit margin.
- To identify high-performing and loss-making products and categories.
- To analyze sales and profit performance across different regions and customer segments.
- To examine monthly sales trends and overall business performance.
- To develop an interactive Power BI dashboard for business reporting and visualization.
- To use DAX measures for business calculations and performance analysis.
- To present analytical findings in a clear and professional format to support business decision-making.

3. DATASET DESCRIPTION

3.1 Dataset Overview

The analysis uses the **Superstore Sample Dataset**, which contains transaction-level retail data covering orders, customers, products, sales, profit, quantity, categories, regions, customer segments, and dates. It was used for analytical and portfolio purposes and does not represent live company transactions.

Dataset Profile	Value
Order lines (rows)	10,194
Fields (columns)	21
Distinct orders	5,111
Distinct customers	804
Distinct products	1,862
Order date range	3 January 2023 to 30 December 2026
Countries	United States (9,994 lines) and Canada (200 lines)
States / provinces	59
Regions	Central, East, South, West
Product hierarchy	3 categories and 17 sub-categories
Customer segments	Consumer, Corporate, Home Office

3.2 Dataset Fields

The 21 fields in the cleaned dataset are listed below.

Field	Description	Type
Row ID	Unique row number for each order line	Number
Order ID	Unique identifier associated with an order	Text
Order Date	Date on which the order was placed	Date
Ship Date	Date on which the order was shipped	Date
Ship Mode	Shipping method used for the order	Text
Customer ID	Unique identifier of the customer	Text
Customer Name	Name of the customer associated with the order	Text
Segment	Customer segment	Text
Country/Region	Country associated with the order	Text
City	City associated with the order	Text
State/Province	State or province associated with the order	Text
Postal Code	Postal code associated with the order	Text
Region	Geographical region	Text
Product ID	Unique identifier of the product	Text
Category	Product category	Text
Sub-Category	Product sub-category	Text
Product Name	Name of the product	Text
Sales	Sales/revenue generated	Number
Quantity	Number of units sold	Number
Discount	Discount rate applied to the order line	Number
Profit	Profit generated	Number

3.3 Supporting Tables

The Excel workbook also contains two supporting tables alongside the main Orders table:

- **Returns** – a list of 296 returned order IDs, which can be joined to Orders by Order ID.
- **People** – the regional manager responsible for each of the four regions.

3.4 Data Usage

The dataset was used across the different stages of the project:

Tool	Role in the Project
Microsoft Excel	Data cleaning, preparation, pivot tables, and KPIs
Microsoft SQL Server	Business analysis and SQL querying
Power BI	Visualization and dashboard development
DAX	Performance measures and business calculations

4. TOOLS & TECHNOLOGIES

4.1 Microsoft Excel

Used for:

- Data cleaning and preparation
- Pivot table analysis
- Initial KPI calculations
- Preliminary data analysis

4.2 Microsoft SQL Server

Used for:

- Data querying, filtering, and sorting
- Aggregation and grouping
- CASE statements for classification
- Subqueries and Common Table Expressions (CTEs)
- Window functions (RANK)
- Business analysis and extracting meaningful insights

4.3 Microsoft Power BI

Used for:

- Data visualization and KPI cards
- Interactive charts, slicers, and filters
- Dashboard development and business reporting

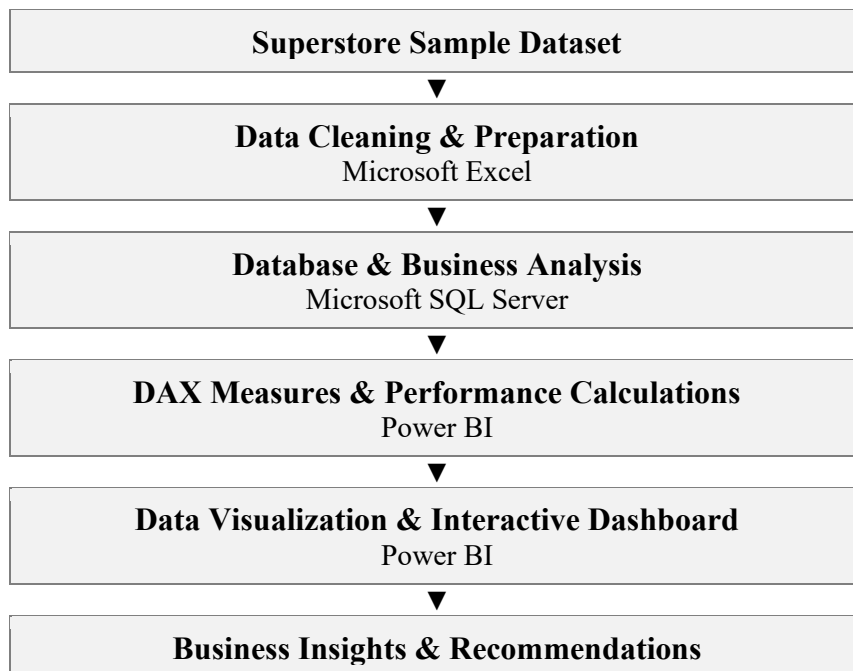
4.4 DAX

DAX was used in Power BI to create measures and perform business calculations, as summarized below.

Measure	DAX Formula	Purpose
Total Sales	SUM(Orders[Sales])	Overall revenue
Total Profit	SUM(Orders[Profit])	Overall profit
Total Quantity	SUM(Orders[Quantity])	Units sold
Total Orders	DISTINCTCOUNT(Orders[Order_ID])	Distinct orders
Profit Margin %	DIVIDE(SUM(Orders[Profit]), SUM(Orders[Sales]), 0)	Profit relative to sales

5. PROJECT WORKFLOW

The project followed an end-to-end analytics workflow, shown below.



Each stage played an important role in transforming raw data into meaningful information for business decision-making.

Overall, the workflow demonstrates how Excel, Microsoft SQL Server, Power BI, and DAX can be combined to prepare data, perform structured analysis, develop an interactive dashboard, and communicate business insights.

6. DATA CLEANING AND EXCEL ANALYSIS

Microsoft Excel was used as the first stage of the project to prepare the Superstore sample dataset and perform preliminary analysis. This stage ensured that the data was properly organized and suitable for further analysis in SQL and Power BI.

6.1 Data Preparation

The dataset was reviewed to understand its structure, columns, records, and data types, then organized so it could be used consistently across Excel, SQL, and Power BI. The main preparation activities were:

- Reviewing the available columns and records.
- Checking the structure of the dataset.
- Converting the **Order Date** column into the appropriate date format.
- Checking numerical fields such as Sales, Profit, and Quantity.
- Identifying and reviewing potential data inconsistencies.
- Preparing the cleaned dataset for SQL analysis.

Validation of the final 10,194 order lines found no missing values, no duplicate rows, and unique Row IDs.

6.2 Pivot Table Analysis

Eight pivot tables were created in Excel to perform an initial analysis of the dataset:

Sheet	Analysis	Measure
PT1	Sales by category	Sum of Sales
PT2	Profit by region	Sum of Profit
PT3	Sales by customer segment	Sum of Sales
PT4	Sales by ship mode	Sum of Sales
PT5	Quantity and sales analysis	Sum of Quantity / Sales
PT6	Monthly sales by year	Sum of Sales
PT7	Top 10 customers by sales	Sum of Sales
PT8	Top 10 products by sales	Sum of Sales

Pivot tables helped summarize a large number of transaction records and provided an initial understanding of business performance before moving to SQL and Power BI.

	A	B	C	D
1				
2				
3	Row Labels	Sum of Sales	Sum of Profit	
4	Furniture	7,54,748	19,730	
5	Office Supplies	7,31,893	1,26,023	
6	Technology	8,39,893	1,46,543	
7	Grand Total	23,26,534	2,92,297	
8				
9				

Excel pivot table – Sales and Profit by Category

6.3 KPI Calculation

Key performance indicators were calculated in Excel for the full dataset and later implemented in Power BI using DAX measures:

KPI	Value	Calculation
Total Sales	4,650,000	Sum of Sales
Total Profit	585,000	Sum of Profit
Total Quantity	77,000	Sum of Quantity
Total Orders	5,000	Distinct count of Order ID
Total Margin	12.57%	Total Profit ÷ Total Sales

6.4 Initial Findings from Excel

The Excel analysis showed that sales and profitability vary widely across regions, states, categories, products, customer segments, and time periods. In particular, Furniture sells almost as much as Technology but earns a small fraction of the profit, and Central earns the lowest regional profit. These patterns became the starting point for the detailed SQL analysis and interactive Power BI reporting.

7. SQL DATA ANALYSIS

7.1 SQL Server Database Setup

After preparing the dataset in Excel, the cleaned data was exported as a CSV file and imported into Microsoft SQL Server. A database named **RetailSalesDB** was created, with the **orders** table storing the cleaned retail transaction data (10,194 order lines). The script starts with a set of checks to confirm the import, shown below with their results on the cleaned dataset.

Check Query	Purpose	Result
SELECT COUNT(*) FROM orders	Row count	10,194
SUM(sales)	Total sales	4,650,000
SUM(profit)	Total profit	585,000
SUM(quantity)	Total quantity	77,000
AVG(discount)	Average discount	15.5%
MAX(sales)	Largest order line	22,638.48
MIN(sales)	Smallest order line	0.44

7.2 SQL Operations Performed

The following SQL concepts were used during the analysis:

- SELECT, WHERE, ORDER BY, and TOP
- GROUP BY and HAVING
- Aggregate functions: SUM(), AVG(), COUNT(), MAX(), and MIN()
- CASE statements for business classification rules
- Window function RANK() OVER for ranking
- Common Table Expressions (CTEs)
- Subqueries (including nested subqueries inside HAVING)

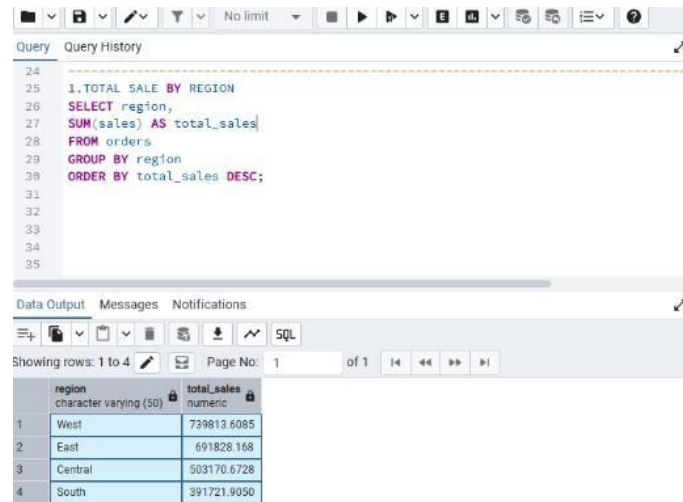
These operations were used to filter, summarize, compare, rank, and classify the transaction data and extract meaningful business insights.

7.3 Sample Queries and Results

Six representative queries are shown below with their results on the cleaned dataset. The complete, runnable script is provided as a separate file.

A. Total sales by region (GROUP BY)

```
SELECT region,
       SUM(sales) AS total_sales
FROM orders
GROUP BY region
ORDER BY total_sales DESC;
```



SQL query and regional sales analysis output

Region	Total Sales	Share of Sales
West	739,814	31.8%
East	691,828	29.7%
Central	503,171	21.6%
South	391,722	16.8%
Total	2,326,534	100.0%

7.4 SQL Analysis Outcome

The SQL analysis provided structured business insights from the transaction-level data. It confirmed the sales and profit patterns seen in Excel and added classifications, rankings, and above-average comparisons that were later represented visually in Power BI.

Working in SQL Server also provided practical experience in using a relational database and writing analytical queries for business decision-making, rather than relying only on spreadsheet-based analysis.

8. POWER BI DASHBOARD DEVELOPMENT

Microsoft Power BI was used to transform the analyzed retail sales data into an interactive business dashboard. The dashboard provides a consolidated view of sales, profit, quantity, orders, and profitability, and allows users to explore the data using interactive filters.

8.1 Data Connection and Preparation

The cleaned retail sales data was connected to Power BI, with the **Orders** table used as the source for the dashboard. The Order_Date field was configured as a date field to support time-based analysis and the monthly sales trend, and all fields were reviewed to ensure they were correctly recognized. DAX measures were then created for the key performance indicators (see Section 8.9).

8.2 Key Performance Indicators

Five KPI cards sit at the top of the dashboard to give an immediate overview of business performance:

- **Total Sales** – the overall sales generated.
- **Total Profit** – the total profit generated.
- **Total Quantity** – the total number of units sold.
- **Total Orders** – the number of distinct orders.
- **Profit Margin %** – total profit as a share of total sales.

8.3 DAX Measures and Calculations

DAX measures provide dynamic calculations that respond to the dashboard filters. The main measures are Total Sales, Total Profit, Total Quantity, Total Orders, and Profit Margin % (see Section 4.4). Two examples:

```
Total Sales = SUM(Orders[Sales])
```

```
Profit Margin % =  
DIVIDE( SUM(Orders[Profit]), SUM(Orders[Sales]), 0 )
```

8.4 Dashboard Design and Formatting

The dashboard was designed as a clean, single-page, business-oriented report. The following design principles were applied:

- Consistent blue and white color theme
- Clear dashboard title and subtitle
- Consistent KPI card and slicer formatting
- Readable chart titles
- Organized visual placement with appropriate spacing
- Interactive filtering for improved usability

8.5 Final Power BI Dashboard

The final dashboard brings together the Excel preparation, SQL analysis, and Power BI visualization stages into a single interactive page that combines KPI cards, sales and profit by

category, top 10 products, monthly trend, regional sales, customer segment distribution, and year and region filters.



Retail Sales & Financial Performance Analysis dashboard

The dashboard in has the Year filter set to 2024 and 2026, so its KPI cards show that selection rather than the full period. The table below compares the two views.

KPI	Dashboard (2024 + 2026)	Full period (2023–2026)
Total Sales	1.22M	2.33M
Total Profit	158.37K	292.30K
Total Quantity	21K	38,654
Total Orders	3K	5,111
Profit Margin %	13.00%	12.56%

9. KEY BUSINESS INSIGHTS

The table below summarizes the main insights and the evidence behind each one.

Sr.no	Insight	Evidence
1	Business is growing and becoming more profitable	Sales up 51% from 2023 to 2026 (494.0K to 745.6K); profit up 86% (51.7K to 95.9K).
2	Furniture is a profitability problem	32.4% of sales, 6.7% of profit, 2.6% margin; Tables lose 17.8K.
3	Deep discounts destroy profit	Lines discounted over 20% lose 136.0K combined; lines over 40% have a -77.4% margin.
4	Central lags the other regions	7.9% margin versus 15.0% in West; average discount of 24.1%.
5	Profit is concentrated in few states	California and New York provide 51.5% of profit; 12 states lose money, led by Texas.
6	Smaller segments are more profitable	Home Office 14.0% and Corporate 13.2% margins versus Consumer 11.6%.
7	Sales are strongly seasonal	September to December produce 51.5% of annual sales.

10. LIMITATIONS AND ASSUMPTIONS

- The Superstore dataset is a sample dataset used for portfolio purposes and does not represent a real company's transactions.
- Monetary values are shown as recorded in the data, without a currency symbol.
- The relationship between discounts and losses is an association observed in the data; it does not by itself prove cause.
- Sales and profit are not adjusted for returned orders.
- The dashboard screenshot reflects the Year filter selection (2024 and 2026); all other figures in Section 9 use the full 2023–2026 period.

11. CONCLUSION

The Retail Sales & Financial Performance Analysis project demonstrates an end-to-end data analytics workflow using Microsoft Excel, SQL, Power BI, and DAX, showing how multiple tools can turn raw retail data into information that supports data-driven decision-making.

Excel was used for data preparation, cleaning, and initial analysis, while SQL was used to run structured queries and extract business insights from the transaction data. Power BI then turned the analyzed data into an interactive dashboard with KPIs, charts, DAX-based calculations, and filters.

The analysis shows a growing business with a 12.6% profit margin, whose profit is held back by deep discounting, an underperforming Furniture category, and weak results in Central and several states. Acting on the recommendations above would address these issues directly.